

1. Which of the following about Routing Protocols is correct?
 - A. RIP 2 is a Distance Vector Protocols
 - B. The RIP 1 uses bandwidth & delay as the metric
 - C. RIP 1 uses subnet mask
 - D. The hold own timer value for RIP is 240 sec
2. Which command will show you all the translations active on your router?
 - A. Show ip nat translations
 - B. Show ip nat statistics
 - C. Debug ip nat
 - D. Clear ip nat translations*
3. The internet is basically known as:
 - A. Network of computers
 - B. Local Area Network
 - C. Network of network
 - D. None of the above
4. The topology with highest reliability is:
 - A. Bus topology
 - B. FDDI
 - C. Star topology
 - D. Mesh topology
5. Which of the following is the way to extract common behavior and attributes from the given classes and make a separate class of those common behaviors and attributes?
 - A. Generalization
 - B. Specialization
 - C. Sub-typing
 - D. Extension
6. Which of the following loop structure ensure execution of loop at least once?
 - A. While
 - B. For
 - C. Do-while
 - D. Nested for
7. Which of the following can access hardware of the computer?
 - A. Structure
 - B. Function
 - C. Pointer
 - D. Union
8. Which of the following features of OOP is used to derive a class from another?
 - A. Encapsulation
 - B. Data hiding
 - C. Polymorphism
 - D. Inheritance
9. There is a unique function in C++ program by where all C++ programs start their execution:
 - A. Start ()
 - B. main ()
 - C. Begin ()
 - D. Output ()
10. Calloc () is used for:
 - A. Static memory management
 - B. Simple memory management
 - C. Dynamic memory management
 - D. None of the above
11. Which of the following is not informed search?
 - A. Hill climbing method
 - B. Beam search method
 - C. Depth first search method
 - D. Tabu method
12. A perception is:
 - A. A multiple layer feed-forward neural network
 - B. A single layer feed-forward neural network with preprocessing
 - C. An auto associative neutral network
 - D. A double layer auto associative neural network

13. A decision problem L is NP-complete if it is in the set of NP problems so that any given solution to the decision problem can be verified in:
 - A. Graph
 - B. Polynomial time
 - C. Array
 - D. Boolean
14. Which of the following is two-way lists?
 - A. Grounded header list
 - B. Circular header list
 - C. Linked list with header and trailer nodes
 - D. None of the above
15. The number of possible binary trees with 4 nodes is:
 - A. 12
 - B. 14
 - C. 13
 - D. 15
16. The complexity of linear search algorithm is:
 - A. $O(n)$
 - B. $O(n^2)$
 - C. $O(\log n)$
 - D. $O(n \log)$
17. Two dimensional arrays are also called:
 - A. Table arrays
 - B. Matrix Array
 - C. Both of the above
 - D. None of the above
18. Binary search algorithm cannot be applied to:
 - A. Sorted linked list
 - B. Sorted linear array
 - C. Sorted binary tree
 - D. Pointer array
19. The order of an algorithm that finds whether a given Boolean Function of 'n' variables produces a 1 is:
 - A. Constant
 - B. Logarithmic
 - C. Linear
 - D. Exponential
20. Propagation delay is the time required for:
 - A. A change in level at the output of an element
 - B. Output to appear when input is applied
 - C. The element becomes functional
 - D. None of the above
21. What name is given to the common microcomputer architecture that used a single block of memory to store both programs and data?
 - A. Von Neumann architecture
 - B. Contemporary architecture
 - C. RISC architecture
 - D. Harvard architecture
22. The MOV M, C instruction of 8085 microprocessor is:
 - A. 1-byte instruction
 - B. 3-byte instruction
 - C. 2-byte instruction
 - D. 4-byte instruction
23. A stack is:
 - A. An 8-bit register in the microprocessor
 - B. An 18-bit register in the microprocessor
 - C. A set of memory locations in R/WM reserved for storing information temporarily during the execution of computer
 - D. A 16-bit memory address stored in the program counter
24. How many address lines would be found on a 128-kbyte memory device (assuming that this is arranged as an array of 8-bit registers)?
 - A. 13
 - B. 17
 - C. 15
 - D. 19

25. How many address lines are needed to address each memory locations in a 2048 x 4 memory chip?
- A. 10 B. 8
C. 11 D. 12
26. Which of the following circuit parameters would be most likely to limit the maximum operating frequency of a flip-flop?
- A. Setup and hold time
B. Propagation delay time
C. Clock pulse HIGH and LOW time
D. Clock transition time
27. An XOR gate produces an output only when its two inputs are:
- A. High
B. Different
C. Low
D. Same
28. The advantage of self-correcting code is that:
- A. It is a weighted code
B. It has even parity
C. It is easy to decode electronically
D. All of the above
29. What is the most widely used method for the automated simplification of Boolean expressions?
- A. Karnaugh maps
B. Fast Fourier transforms
C. Quine-McCluskey minimization
D. Binary reduction
30. The bit sequence 0010 is serially entered (right-most bit first) into a 4-bit parallel out shift register that is initially clear. What are the Q outputs after two clock pulses?
- A. 0000 B. 1000
C. 0010 D. 1111
31. Which of the following might be output as a result of using CASE tools?
- A. Prototypes
B. Programming codes
C. Cost analysis
D. All of the above
32. The incremental model of software development is:
- A. A reasonable approach when requirements are well defined
B. A good approach when a working core product is required quickly
C. The best approach to use for projects with large development teams
D. A revolutionary model that is not used for commercial products
33. How many steps are in the systems development life cycle (SDLC)?
- A. 4 B. 6
C. 5 D. 10
34. The application development without programming structured system analysis is:
- A. Not useful at all
B. Of less use than in application development with programming
C. Usually done after system design
D. Heavily used
35. The high degree of Modularity of Software Component is achieved by having
- A. High coupling and high cohesion
B. High coupling and low cohesion
C. Low coupling and high cohesion
D. Low coupling and low cohesion
36. Which is not the common DFD drafting errors?
- A. Grey hole B. Black hole
C. Green hole D. Miracle

37. The linear sequential model of software development is also known as the:
- A. Fountain Model
 - B. Spiral Model
 - C. Classical Life Cycle Model
 - D. Chaos Model
38. Which of the following are characteristics of an RDBMS?
- A. Queries are possible on individual or groups of tables
 - B. Keys may be unique or have multiple occurrences in the data base
 - C. Data are organized in a series of two-dimensional tables each of which contains records for one entity
 - D. Tables are linked by common data known as keys
39. Which of the following is not a logical data-base structure?
- A. Tree
 - B. Chain
 - C. Relational
 - D. Network
40. A relation is in this form if it is in BCNF and has no multi valued dependencies:
- A. Second normal form
 - B. Fourth normal form
 - C. Third normal form
 - D. Domain/key normal form
41. Which of the following is a problem of file management system?
- A. Difficult to update
 - B. Program dependence
 - C. Data redundancy
 - D. All of the above
42. Assume transaction A holds a lock R, if transaction B also request for a shared lock on R, it will:
- A. Result in deadlock situation
 - B. Immediately be granted
 - C. Immediately be rejected
 - D. Be granted as soon as it is released by A
43. The DBMS language component which can be embedded in a program is:
- A. Data definition language (DDL)
 - B. The data base administrator (DBA)
 - C. Data manipulation language (DML)
 - D. A query language
44. A critical region:
- A. Is a piece of code which only one process executes at a time?
 - B. Is a region prone to deadlock?
 - C. Is a piece of code which only a finite number of processes execute
 - D. Is found only in Windows NT operation system
45. Choose the correct statements:
- A. A disk cannot have more than one file system stored in it.
 - B. The logical to physical device address mapping is done by the device driver.
 - C. On the logical level, the kernel deals with disks rather than file system.
 - D. None of the above
46. In the blocked state:
- A. The processes waiting for I/O are found
 - B. The process which is running is found
 - C. The processes waiting for the processor are found
 - D. None of the above

47. Thrashing is the computer activity which:

- A. Reduces page I/O
- B. Decreases the degree of multiprogramming
- C. Implies excessive page I/O
- D. Improve the system performance.

48. A page fault occurs:

- A. When the page is in the memory
- B. When the process enters the blocked state
- C. When the page is not in the memory
- D. When the process is in the ready state

ANSWER SHEET

1.A	2.B	3.C	4.D	5.A	6.C	7.C	8.D	9.B	10.C
11.C	12.D	13.B	14.D	15.B	16.A	17.C	18.A	19.D	20.B
21.A	22.C	23.C	24.B	25.C	26.B	27.B	28.C	29.C	30.B
31.D	32.B	33.B	34.B	35.C	36.D	37.C	38.D	39.B	40.B
41.D	42.B	43.C	44.A	45.B	46.A	47.D	48.C		